

# Certification of Imitation Controllers Using Learned Lyapunov Functions

Master thesis by Josua Christoph Lindemann

Date: June 23, 2026, Date of submission: 16. August 2026

1. Review: Prof. Dr.-Ing. Rolf Findeisen

2. Review: Hendrik Alsmeier M.Sc.

Darmstadt



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Control and Cyber-Physical Systems

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Information Technology  
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# 1 Abstract

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## 2 Introduction

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## 3 Optimal Control and Lyapunov Stability Theory

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### 3.1 Discrete-Time Dynamical Systems and Stability

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To implement an optimal control scheme on digital hardware, the continuous-time system dynamics must be discretized. Therefore, a general nonlinear system is considered to be:

$$\dot{x}(t) = f(x(t), u(t)) \quad (3.1)$$

where  $x(t) \in \mathcal{X}$  and  $u(t) \in \mathcal{U}$  are the constrained state and input vectors, respectively.

TODO: MISSING TEXT

$$x_{k+1} = x_k + T_s f(x_k, u_k) \quad (3.2)$$

$$x_{k+1} = x_k + \frac{T_s}{6}(k_1 + 2k_2 + 2k_3 + k_4), \quad (3.3)$$

where the intermediate slopes  $k_1$  through  $k_4$  are defined as:

$$\begin{aligned} k_1 &= f(x_k, u_k) \\ k_2 &= f\left(x_k + \frac{T_s}{2}k_1, u_k\right) \\ k_3 &= f\left(x_k + \frac{T_s}{2}k_2, u_k\right) \\ k_4 &= f(x_k + T_s k_3, u_k). \end{aligned} \quad (3.4)$$

In the following, the resulting discrete-time system can be expressed in the general mapping form:

$$x_{k+1} = f_d(x_k, u_k). \quad (3.5)$$

#### 3.1.1 Lyapunov Stability

Let the origin  $(x, u) = (\mathbf{0}, \mathbf{0})$  be an equilibrium point of the system such that  $f_d(\mathbf{0}, \mathbf{0}) = \mathbf{0}$ . Suppose the system is controlled by a state-feedback control policy  $u_k = \kappa(x_k)$ . The closed-loop system is said to be *Lyapunov stable* if there exists a positive definite Lyapunov function  $V(x_k)$  such that:

$$\begin{aligned} V(\mathbf{0}) &= 0 \\ V(x_k) &> 0 \quad \forall x_k \in \mathcal{X} \setminus \{\mathbf{0}\} \\ V(f_d(x_k, \kappa(x_k))) - V(x_k) &\leq 0 \quad \forall x_k \in \mathcal{X} \setminus \{\mathbf{0}\}. \end{aligned} \quad (3.6)$$

Furthermore, the equilibrium point is *locally asymptotically stable* if the Lyapunov function additionally satisfies a strictly negative definite decrease condition along the closed-loop trajectories:

$$V(f_d(x_k, \kappa(x_k))) - V(x_k) < 0 \quad \forall x_k \in \mathcal{X} \setminus \{\mathbf{0}\}. \quad (3.7)$$

Asymptotic stability is typically the desired property for control systems, as it ensures that the system will converge to the equilibrium point over time.

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## 3.2 Model Predictive Control and Stability

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### 3.2.1 Problem Formulation

$$\begin{aligned} \{x_k^*\}_0^N, \{u_k^*\}_0^{N-1} = & \underset{\{x_k\}_0^N, \{u_k\}_0^{N-1}}{\operatorname{argmin}} \sum_{k=0}^{N-1} \ell(x_k, u_k) + V(x_N) \\ \text{s.t. } & x_{k+1} = f(x_k, u_k) & \forall k \in [0, \dots, N-1] \\ & x_k \in \mathcal{X}_k & \forall k \in [0, \dots, N] \\ & u_k \in \mathcal{U}_k & \forall k \in [0, \dots, N-1] \\ & x_0 = x_{init}. \end{aligned} \quad (3.8)$$

### 3.2.2 Terminal Set and Terminal Cost Constraints

### 3.2.3 Closed-Loop Stability

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## 3.3 Stability without Terminal Constraints

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## **4 Foundations of Neural Control Policies**

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### **4.1 Artificial Neural Networks**

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#### **4.1.1 Multi-Layer Perceptrons**

#### **4.1.2 Optimization and Training**

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### **4.2 Counterexample-Guided Inductive Synthesis**

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## 5 Formal Verification of Neural Control Policies

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### 5.1 Mixed-Integer Linear Programming

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### 5.2 Branch and Bound

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### 5.3 $\alpha\beta$ -CROWN

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## **6 Learning Lyapunov-Stable Imitation Controllers**

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### **6.1 Neural Policy Approximation and Imitation Learning**

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#### **6.1.1 Data Generation and MPC Expert**

#### **6.1.2 Policy Network Architecture and Imitation Objective**

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### **6.2 Lyapunov Learning for Stability Certification**

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#### **6.2.1 Lyapunov Network Architecture**

#### **6.2.2 Loss Formulation and CEGIS Training**

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### **6.3 Formal Verification Methodology**

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## 7 Experimental Evaluation and Certification Results

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### 7.1 System Modeling and Problem Formulation

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For our experimental setup, two benchmark systems are selected: the double integrator and the cartpole. These systems provide a good balance between simplicity and complexity allowing us to certify the stability of the closed-loop system.

#### 7.1.1 Double Integrator

The double integrator is a simple second order linear system, therefore it serves as the ideal baseline system in the learning and certification pipeline. The continuous-time dynamics are described by the following state-space equation:

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \quad (7.1)$$

where  $x(t) = [p(t) \ v(t)]^\top$  is the state vector containing the position and velocity of the system, respectively, and  $u(t)$  denotes the scalar control input (acceleration).

For the *discretization* of the continuous-time system, either Euler's method or the RK4 method with a sampling time of  $T_s = 0.1$  s is chosen, depending on certification or simulation purposes. Applying Euler's method (Eq. (3.2)) yields the discrete-time dynamics  $x_{k+1} = f_{\text{Euler}}(x_k, u_k)$ , whereas the RK4 method (Eq. (3.3)) results in  $x_{k+1} = f_{\text{RK4}}(x_k, u_k)$ .

The *MPC setup* is formulated according to the problem in Eq. (3.8) subject to the double-integrator dynamics in Eq. (7.1). A prediction horizon of  $N = 20$  steps is chosen, and the states and inputs are penalized using the weighting matrices  $Q = \text{diag}(15, 1)$  and  $R = 0.1$ .

#### 7.1.2 Cartpole

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### 7.2 Linear System: Double Integrator

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### 7.3 Nonlinear System: Cartpole

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## 8 Discussion

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### 8.1 Qualitative Discussion of Results

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### 8.2 Challenges with Nonlinear Dynamics

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### 8.3 Scalability and Limitations

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## 9 Conclusion

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## A Appendix

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